

- 1 A student planned to investigate the rate of respiration in germinating seeds and then determine the respiratory quotient (RQ) of the germinating seeds.

The student used the respirometer shown in Figure 1.1 for the investigation.

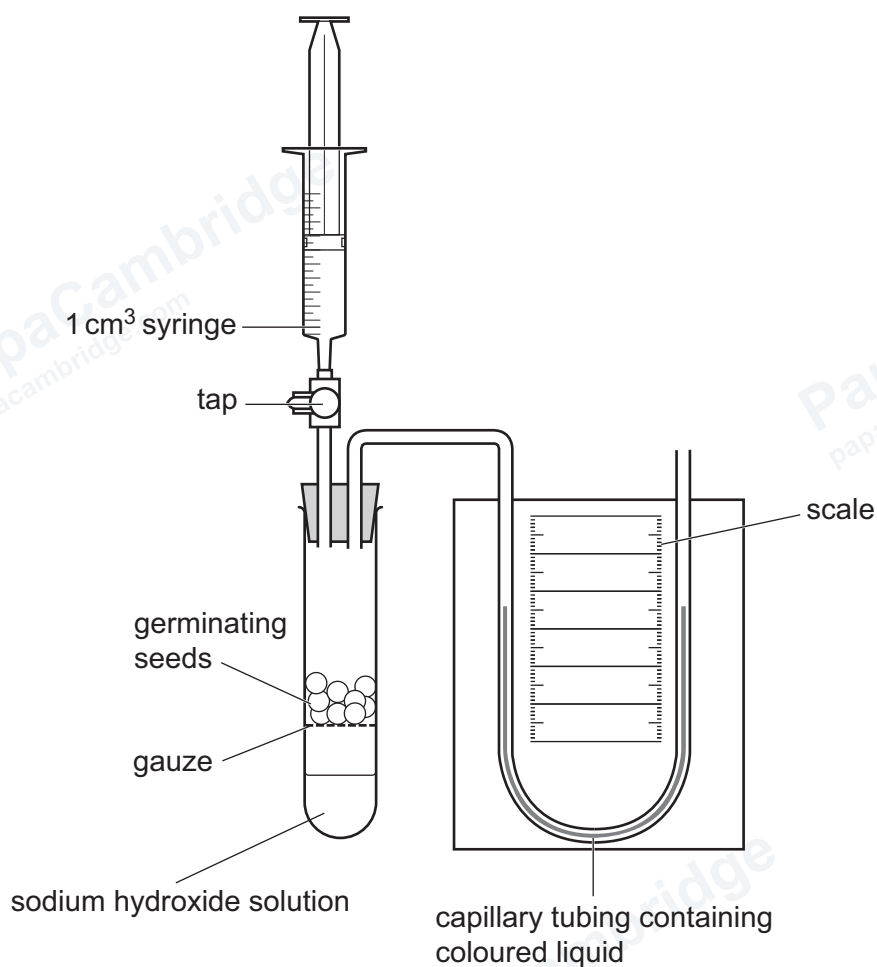


Figure 1.1

- (a) The student investigated the effect of temperature on the rate of respiration in the germinating seeds.

The student planned to use a temperature range of 10–50 °C in the investigation.

The student read that in some species:

- at lower temperatures, the rate of respiration approximately doubles for each 10 °C increase in temperature
 - the optimum temperature for respiration is approximately 40 °C.
- (i) Sketch a curve on Figure 1.2 to show the results that you predict the student will produce from this investigation.

Include labelled axes. Do **not** include units or values on the axes.

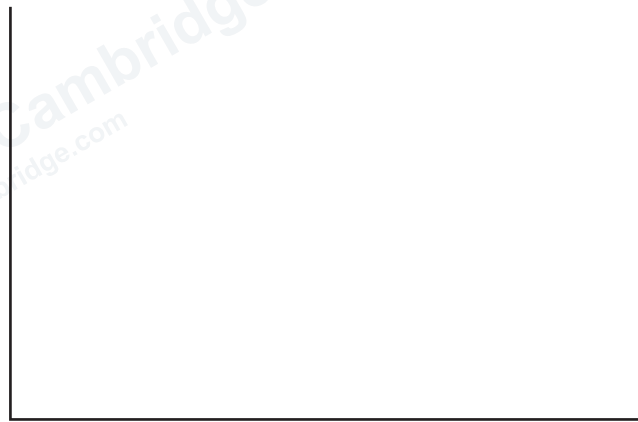
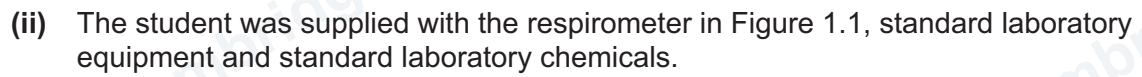


Figure 1.2

[3]



The description of your method should be set out in a logical way and be detailed enough for another person to follow.

[8]



- (iii) The diameter of the capillary tubing shown in Figure 1.1 was 1 mm.

The student used this value and the measurements that were taken to calculate a rate of respiration at each temperature.

Describe each step in the calculation that the student used to determine a rate of respiration.

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..... [4]

- (b) The student then decided to determine the RQ of germinating seeds from four different species of plant. The seeds of all the species are known to respire aerobically.

For each species of plant, the student placed the germinating seeds in the respirometer and then measured the volume of oxygen used and the volume of carbon dioxide produced in one minute. The student decided to carry out all the measurements at 30 °C.

The results are shown in Table 1.1.

Table 1.1

species of plant	mean volume of oxygen used in one minute/mm ³	mean volume of carbon dioxide produced in one minute/mm ³
A	0.065	0.055
B	0.056	0.059
C	0.027	0.021
D	0.052	0.045

- (i) Use Table 1.1 to calculate the RQ of species **A**.

RQ = [1]

- (ii) The student suggested that the results for one of the species were **not** as expected and the procedure should be repeated for that species.

With reference to Table 1.1, state the species, **A**, **B**, **C** or **D**, for which the procedure should be repeated. Give a reason for your choice.

species

reason

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..... [2]

- (iii) Suggest an error in the procedure that could explain the results for the species you stated in **1(b)(ii)**.

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..... [1]

[Total: 19]